| Tech Management, SRI-B

CLASH OF THE

CLAWS

MS Ramaiah Institute of Technology _DevMind

Your Codebase's Strategic Twin

Productivity Platforms | Repo Intelligence + Sovereign AI

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Problem Statement

Theme: Productivity Platforms — software delivery is bottlenecked by communication, not just coding.

Problem (The Context Gap): Technical execution (PRs/Commits) vs. business understanding.

- Docs are always outdated; handover is manual and painful.
- Developers spend 68% of their time reading code instead of writing it.
- A month of productivity is lost every time a new hire joins.

Who Faces It:

- The New Hire — needs to understand a 50k-line codebase in hours, not weeks.
- The PM — needs plain-English summaries of what shipped.
- The CTO — needs a high-level view of architecture health & deployment readiness.

Why It Matters: Turns the repo from a black box into a living, explaining entity.

- Onboarding drops from weeks to hours.
- 68% dev time reading code; 2–4 weeks lost during dev turnover on undocumented projects.

Current Solution & Gaps

Existing Tools: GitHub PR Reviews, Notion, Confluence, Jira, and basic AI coding assistants.
Strengths: Great for version control (GitHub) or static documentation (Notion).

What's Missing / Broken:

- Docs in Notion/Confluence are static — they die the moment code changes.
- Humans still manually write release notes and technical debt reports.
- GitHub PRs are written by devs for devs; unreadable to business stakeholders.
- PMs have no automated way to see the 'business impact' of a technical refactor.
- Most AI tools require sending your entire codebase to the cloud — a massive security risk.

Opportunity:

- Market Gap: A 'Living Documentation' system that updates itself with every merge.
- LLMs are perfect for translating complex code diffs into human-readable strategic summaries.
- Rise of powerful local LLMs (Llama 3) makes this possible securely — no cloud needed.
- 'Sovereign Intelligence': the AI lives inside the repo.

Idea

Your Solution

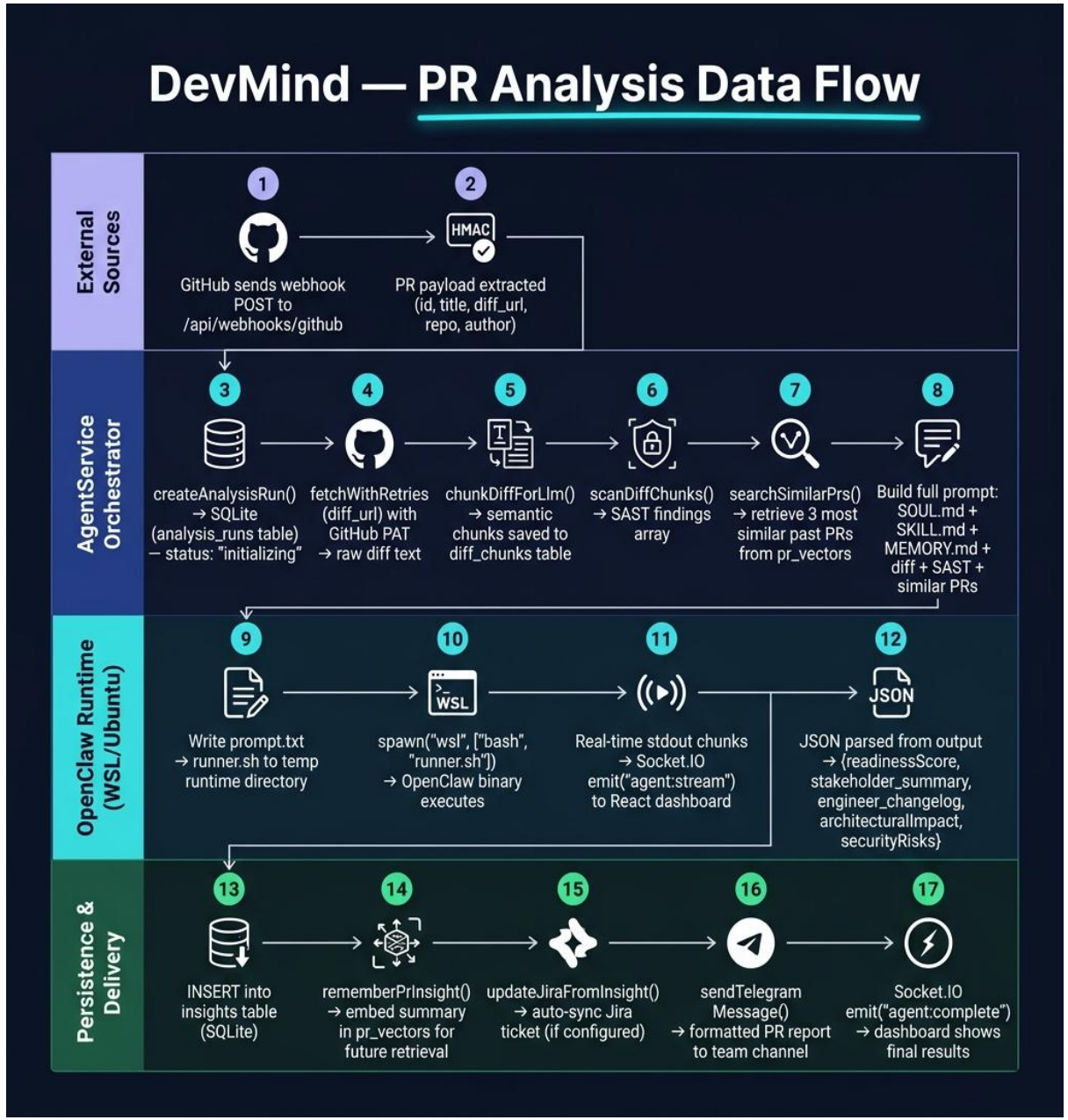
What We Built: A local-first PR intelligence and repository mapping platform. DevMind translates complex code changes into plain-English business insights while mapping
your entire architecture in real-time. An AI that explains your codebase to everyone — devs to CEOs.

Core Idea: Acts as a 'Strategic Twin' — monitors every change, updates its mental model of the project.
Uses OpenClaw (local AI stack) to analyze PR diffs & repo structures — no code leaves the machine.

Key Concepts:
- Repo X-Ray — deep structural scan of files and dependencies.
- Readiness Score — metric-driven confidence score for deployment.
- Architecture Mapper — visual graph of how modules interact.

Key Features: Repo X-Ray | Auto Handover Docs | Live ChangeLog Engine | Business Insight Translator
Readiness Score | Architecture Mapper | Dual Persona UI | Sovereign AI Core

Strongest Innovation: Sovereign AI Core — deep code reasoning entirely on-device.
Most Value: Business Insight Translator — saves non-technical leaders hours of 'what changed?' meetings.

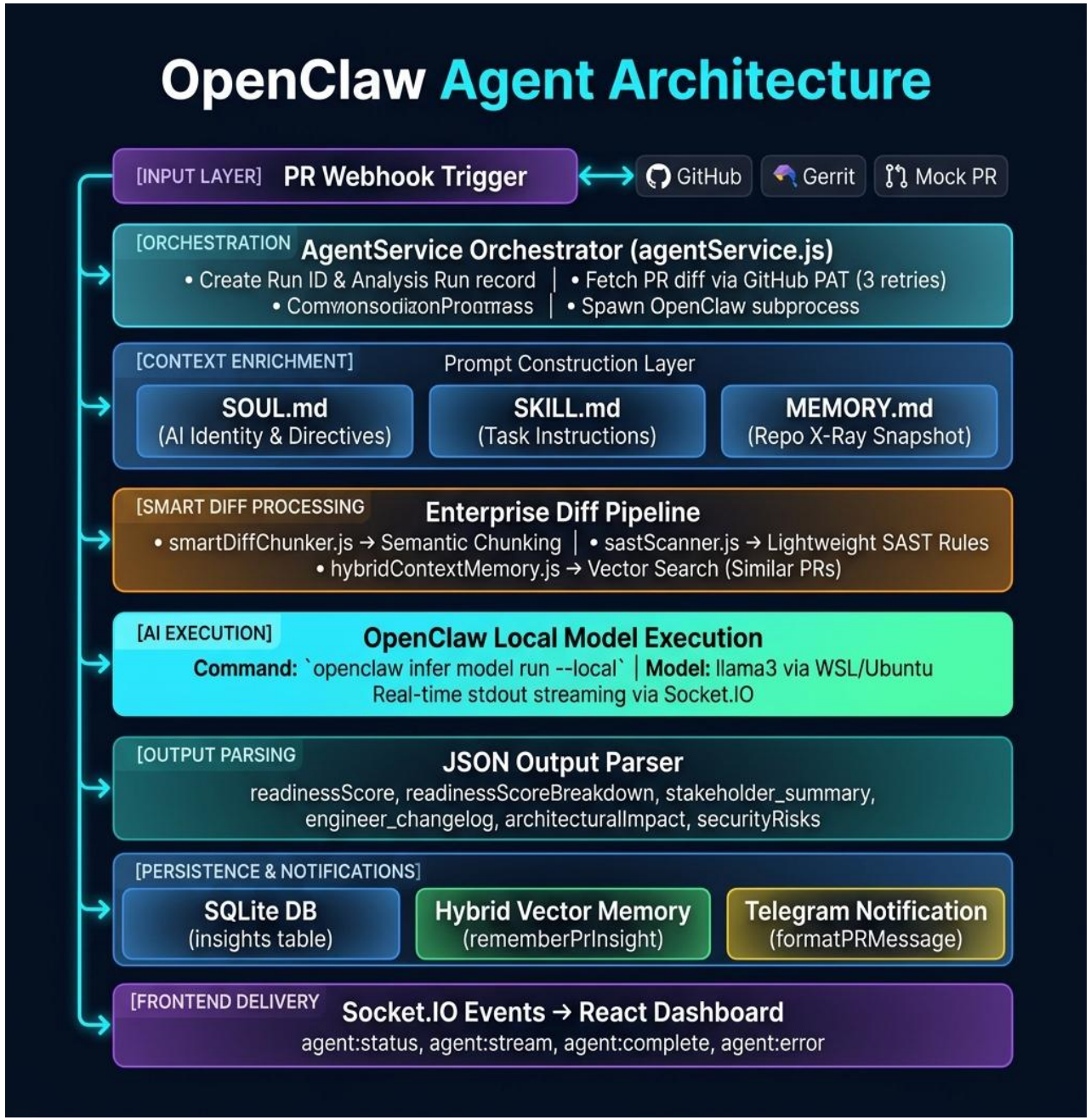


Demo / Product Walkthrough

Flow: GitHub PR Merged → DevMind Server (Local) → AI Processing → Dashboard + Mobile Notifications

Step 1 — Trigger: A developer merges a Pull Request on GitHub.
Step 2 — Webhook: GitHub sends a webhook to the DevMind Backend.
Step 3 — Ingestion: System clones/fetches the diff and queues it for analysis.
Step 4 — AI Processing: Local AI (Llama 3 via OpenClaw) analyzes the diff using:
SOUL (strategy) | SKILL (logic) | MEMORY (context) modules.
Step 5 — Generation: Two outputs created:
- Technical ChangeLog — for developers.
- Business Summary — for Product Managers.

Step 6 — Reporting: Readiness Score calculated (security risks + architectural impact).
Step 7 — Delivery: Dashboard updates via Socket.IO; summary sent via Telegram / WhatsApp.

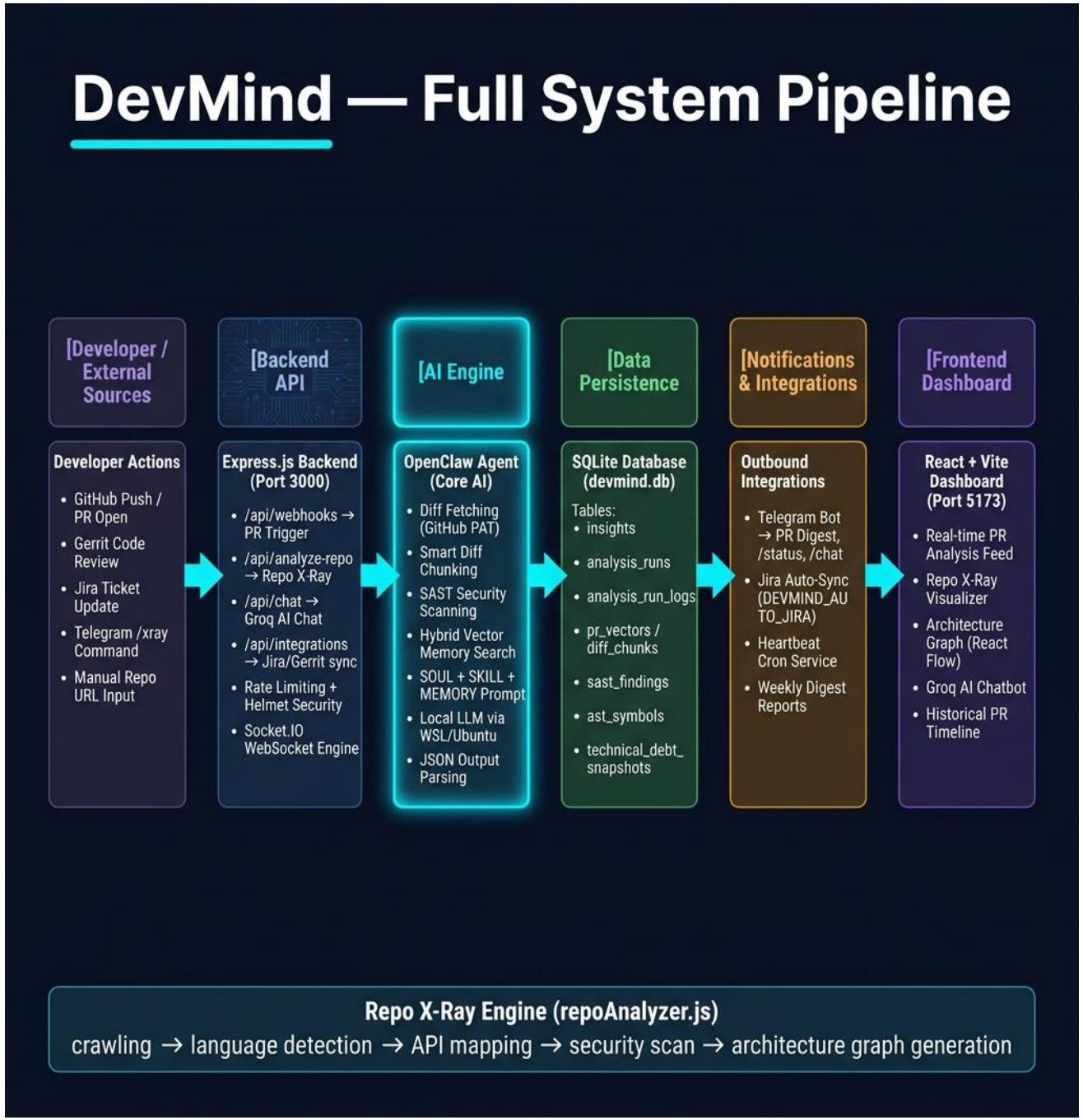


Tech & Architecture

Tech Stack:
- Frontend: React, Vite, Tailwind CSS, React Flow
- Backend: Node.js, Express
- Database: SQLite (local-first persistence)
- Real-time: Socket.IO — live streaming logs and UI updates
- Security: GitHub Webhook HMAC verification + local-only processing

AI Usage:
- Model: Llama 3 (8B) — best-in-class for local execution
- Frameworks: OpenClaw Stack | Ollama | Groq (metadata-only cloud fallback)
- Local vs Cloud: Primary intelligence is LOCAL (Sovereign); only sanitized metadata to cloud
- Roles: AST parser (structure) | Logic summarizer | Risk calculator

System Architecture:
Webhook → Express → OpenClaw (Local AI) → SQLite / Socket.IO
Layers: Ingestion → Processing (AI) → Persistence (DB) → Delivery (UI / Mobile)
Integrations: GitHub | Telegram | WhatsApp
Security: Code never leaves local server; only plain-English summaries sent to Telegram.

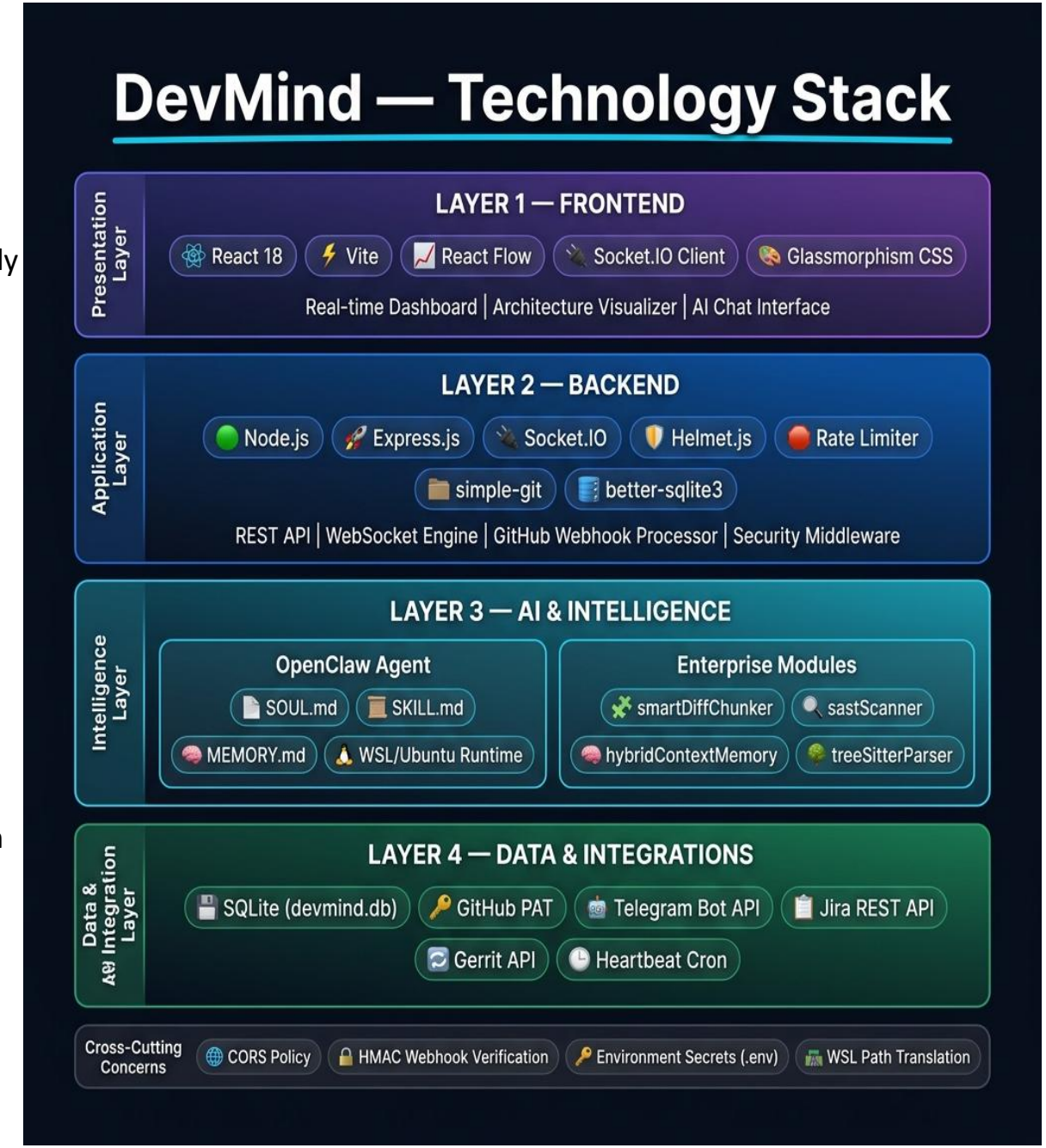


Impact & Use Cases

Real-World Impact:
- Productivity: Reduces standup meeting times by 40% — everyone is already informed.
- Onboarding: A new dev can 'chat' with the repo to understand why a feature was built.
- Management: Non-technical founders stay in the loop without bothering developers.
- Deployment: Readiness Score identifies blockers before they hit production.
- Documentation: 100% automated, always-up-to-date documentation.

Use Cases:
- The Handover: Senior dev leaves → new hire uses Auto Handover Docs — learns system in 1 day.
- The Weekly Sync: PM receives a Friday WhatsApp digest of everything shipped that week.
- The Security Audit: Architecture Mapper flags a new unexpected connection to an external API.

Personas:
Devs → ChangeLog | PMs → Business Insight Translator | CTOs → Architecture Map



Differentiation & Competitive Advantage

What Makes Us Unique:

- We don't just 'chat with code' — we translate code into business strategy.
- Privacy-first: Enterprises can't use ChatGPT on private repos. They CAN use DevMind.
- Dual Persona UI — separate views for engineers and business owners.
- Technical Advantage: Local LLMs + static repo analysis (AST parsing) combined.

Strong MOAT (Sustainable Competitive Advantages):

1. Sovereign AI: Local-first processing makes it 'Audit-proof.'
2. Real-time Business Summaries: No one else translates PRs into PM-friendly language automatically.
3. Privacy Guarantee: Cloud-based competitors (GitHub Copilot) cannot guarantee code privacy.
4. Project Memory: Gets smarter with every PR merged — long-term institutional knowledge.

In Business — A moat is a sustainable competitive advantage that protects a company from competitors and helps maintain long-term profitability.

Privacy, Security & Sovereignty

Source Code Protection:

- All raw code, diffs, and AST trees stay on the local workstation or private server.
- Zero raw code ever goes to the cloud.
- Only sanitized metadata (when using Groq) or notification summaries are sent externally.

Security Mechanisms:

- HMAC Webhook Signatures — every GitHub webhook is cryptographically verified.
- Environment-based secret management — no credentials in code.

Sovereignty:

- By using OpenClaw, the AI model becomes a 'Sovereign Asset' of the company.
- Not a rented service — it lives inside your infrastructure.

GitHub Repo: https://github.com/ayk7115/DevMinds.git
Demo Link: https://youtu.be/IzbLwM5gIj4
Document: Submitted ✓

Idea

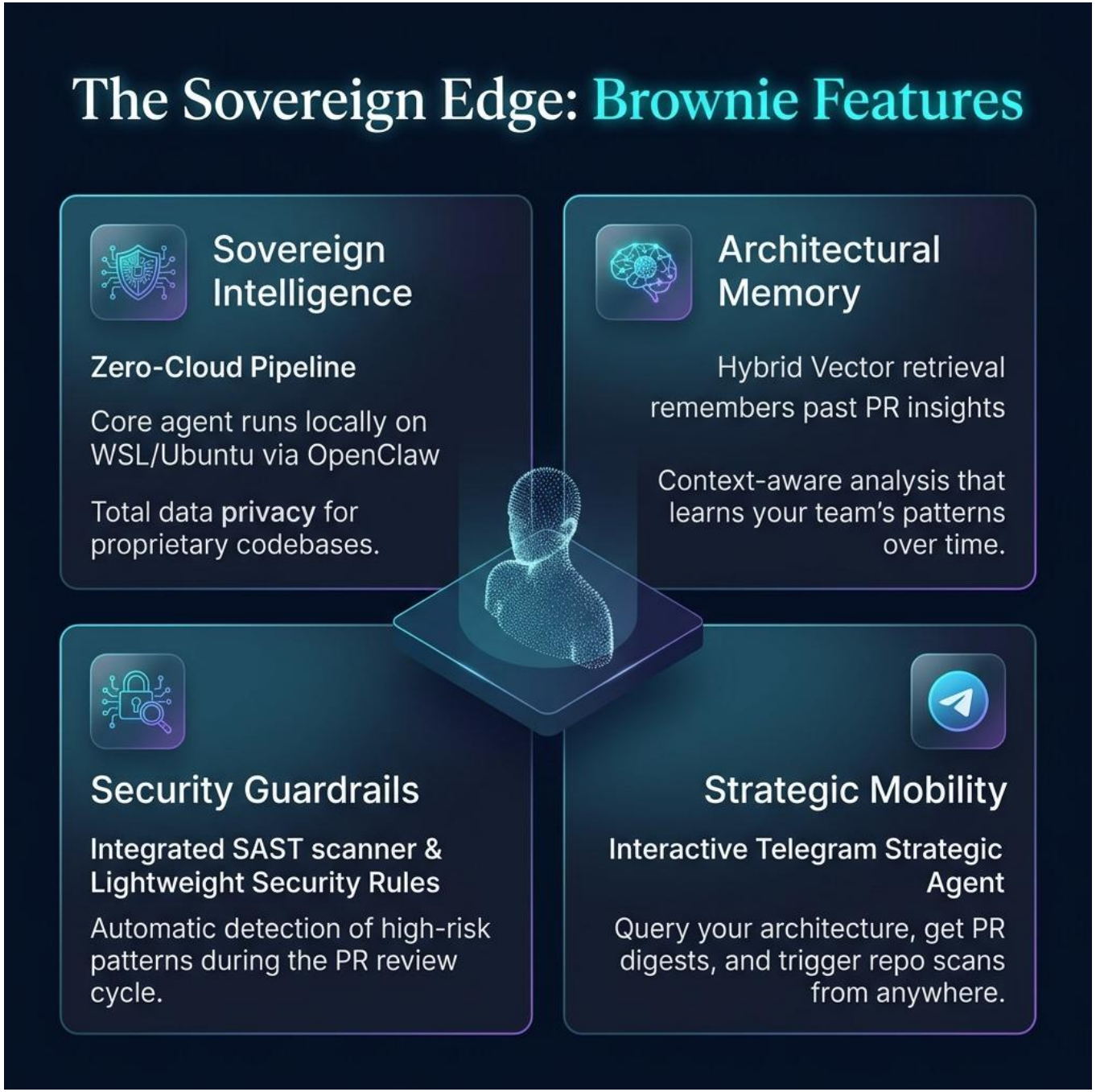
Brownie

Key Metrics:
- 50% faster onboarding
- 100% documentation coverage
- 0% code leakage

Core Benefits:
- Clarity — stakeholders always understand what shipped and why.
- Continuity — engineering knowledge survives team changes.
- Security — enterprise-grade privacy with no cloud dependency.

Adoption: Slots into existing GitHub workflows with zero friction.

Future-Ready: As codebases grow with AI-generated code, human-readable summaries become the only way to manage the volume of changes.

Roadmap: Future support for Gerrit | Jira automation | Technical debt trend prediction.

"DevMind: Translating the pulse of your code into the language of your business."